SRM Institute of Science and Technology
College of Engineering and Technology
School of Computing
SRM Nagar, Kattankulathur – 603203, Chengalpattu District, Tamilnadu

Academic Year: 2025-26 (ODD)

Test: FT II
Course Code & Title: 21CSS201T/Computer Organization and Architecture
Year & Sem: II Year / III Sem

Date: 15.09.25
Duration: 2 Hours
Max. Marks: 50

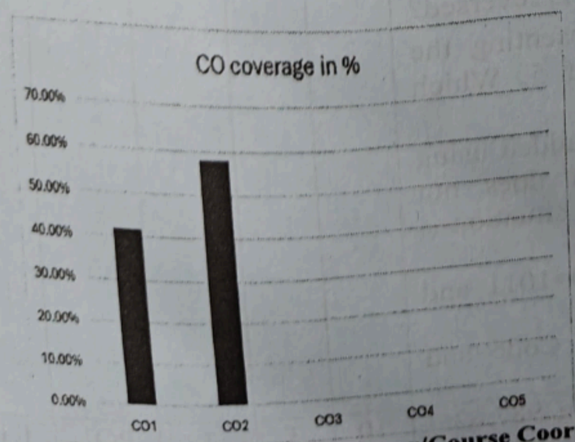
Part – B
(4 x 10 =40 Marks)
Answer any Four

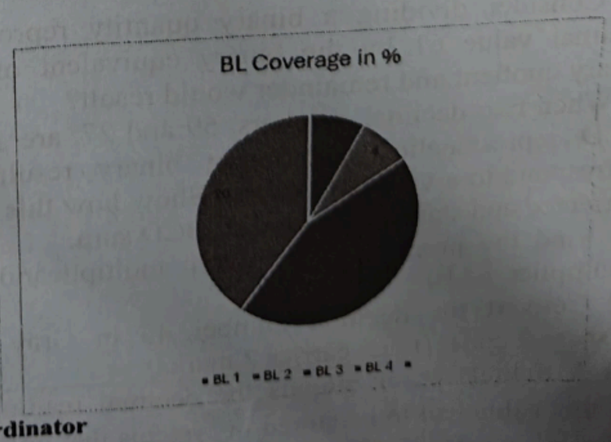
11	a) A system uses 8-bit 2's complement arithmetic. • If one register holds the value 75 and another holds 48, find the difference. • What happens if the order of subtraction is reversed? b) Consider dividing a binary quantity representing the decimal value 61 by the binary equivalent of 5. Which binary quotient and remainder would result? c) When two decimal numbers, 59 and 27, are added using BCD representation, the direct binary result does not correspond to a valid BCD digit. Show how this situation is corrected and state the final valid BCD sum. d) Find the product of the given multiplicand=1011 and multiplier = 110 e) Convert the decimal number 45 in Gray Code and Excess-3 code. (Each carries 2 marks)	10	3	1	PO1	1.1.2
12	a) A digital sensor outputs the decimal reading 987.625. If this value had to be stored in systems that only recognize binary, octal, or hexadecimal formats, how could you represent it in each of those bases? Outline the reasoning for every step you take. (3 marks) b) An engineer encounters the binary fraction 1011011.101. Convert binary fraction to decimal and Hexadecimal values. (3 marks) c)A dataset includes the octal code 7365. Before using it in a binary-compatible circuit, and later in a hexadecimal-coded system, the number must first be expanded into binary and then regrouped for hexadecimal. (2 marks) d) Convert the hexadecimal number $(2F7A)_{16}$ into binary and then into decimal. (2 marks)	10	3	1	PO1	1.1.2
13	a) In the architecture of the 8086 microprocessor, discuss the role of the following components: (5 marks) • Instruction Queue • Segment Registers • Flag Register	10	4	2	PO2	2.2.1

studocu
messages.pdf_cover_qr_code_label
messages.downloaded_by

	b) With the help of a neat labelled diagram, draw the architecture of the 8086, clearly showing the Bus Interface Unit (BIU) and Execution Unit (EU). (5 marks)					
14	List and explain addressing modes of the 8086 microprocessor with suitable examples. For instance, consider the following instructions and identify the addressing modes used: i) MOV AX, [BX + SI] (3 marks) ii) MOV AL, [BX + 0050H] (4 marks) iii) MOV AX, [SI] (3 marks)	10	4	2	PO2	
15	a)Compare single-bus and multi-bus organizations in computer architecture. Highlight their advantages and disadvantages with neat diagrams. (5 Marks) b) Discuss the memory hierarchy and Memory Locations and Addresses. (5 Marks)	10	4	2	PO2	2

Course Outcome (CO) and Bloom's level (BL) Coverage in Questions





Approved by the Audit Professor/Course Coordinator

studocu
messages.pdf_cover_qr_code_label
messages.downloaded_by